| 3 | **Identities and inequalities** | **Notes** |
|---|---|---|
| | Simple algebraic division. | Division by $(x + a)$, $(x - a)$, $(ax + b)$ or $(ax - b)$ will be required. |
| | The factor and remainder theorems. | Students should know that if $f(x) = 0$ when $x = a$, then $(x - a)$ is a factor of $f(x)$.<br>Students may be required to factorise cubic expressions such as: $x^3 + 3x^2 - 4$ and $6x^3 + 11x^2 - x - 6$, when a factor has been provided.<br>Students should be familiar with the terms 'quotient' and 'remainder' and be able to determine the remainder when the polynomial $f(x)$ is divided by $(ax + b)$ or $(ax - b)$. |
| | Solutions of equations, extended to include the simultaneous solution of one linear and one quadratic equation in two variables. | The solution of a cubic equation containing at least one rational root may be set. |
| | Simple inequalities, linear and quadratic. | For example $ax + b > cx + d$, $px^2 + qx + r < sx^2 + tx + u$. |
| | The graphical representation of linear inequalities in two variables. | The emphasis will be on simple questions designed to test fundamental principles.<br>Simple problems on linear programming may be set. |